

Automata and Formal Languages

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Regular Expressions

- *Regular expressions* are an algebraic way to describe languages.
- They describe exactly the regular languages.
- If E is a regular expression, then $L(E)$ is the language it defines.
- We'll describe RE's and their languages recursively.

The Operators of Regular Expressions

- The **Union** of two languages L and M , denoted $L \cup M$, is the set of strings that are in either L or M , or both.
 - For example, if $L = \{001, 10, 111\}$ and $M = \{\epsilon, 001\}$ then $L \cup M = \{\epsilon, 001, 10, 111\}$
- The **Concatenation** of languages L and M is the set of strings that can be formed by taking any string in L and concatenating it with any string in M .
 - For example, if $L = \{001, 10, 111\}$ and $M = \{\epsilon, 001\}$ then $L.M = \{001, 10, 111, 001001, 10001, 111001\}$

The Operators of Regular Expressions

- The **Closure** (or star or Kleene closure) of a language L is denoted L^* represents the set of those strings that can be formed by taking any number of strings from L .
- If $L = \{0, 1\}$, then L^* denotes all strings of 0's and 1's.
- Note: Formally, $L^* = \bigcup_{i \geq 0} L^i$ which is an infinite set
- Can you give an example of a language whose closure is not infinite?
 - $\phi^* = \{\epsilon\}$ as $\phi^0 = \{\epsilon\}$ and ϕ^i is empty for $i \geq 1$

Building Regular Expressions

- **Basis 1:** If 'a' is any symbol, then **a** is a RE, and $L(\mathbf{a}) = \{a\}$.
 - **Note:** $\{a\}$ is the language containing one string, and that string is of length 1.
- **Basis 2:** ϵ is a RE, and $L(\epsilon) = \{\epsilon\}$.
- **Basis 3:** $\emptyset$ is a RE, and $L(\emptyset) = \emptyset$.

Building Regular Expressions

- **Induction 1:** If E_1 and E_2 are regular expressions, then $E_1 + E_2$ is a regular expression, and $L(E_1 + E_2) = L(E_1) \cup L(E_2)$.
- **Induction 2:** If E_1 and E_2 are regular expressions, then $E_1 E_2$ is a regular expression, and $L(E_1 E_2) = L(E_1) L(E_2)$.
 - **Concatenation** : the set of strings wx such that w is in $L(E_1)$ and x is in $L(E_2)$.
- **Induction 3:** If E is a RE, then E^* is a RE, and $L(E^*) = (L(E))^*$.
 - **Closure**, or "Kleene closure" = set of strings $w_1 w_2 \dots w_n$, for some $n \geq 0$, where each w_i is in $L(E)$.
 - **Note:** when $n=0$, the string is ϵ .

Precedence of Operators

- Parentheses may be used wherever needed to influence the grouping of operators.
- Order of precedence is * (highest), then concatenation, then + (lowest).
- Examples:
 - $L(\mathbf{01} + \mathbf{0}) = \{01, 0\}$.
 - $L(\mathbf{0}(\mathbf{1} + \mathbf{0})) = \{01, 00\}$.

Regular expressions

- Identities & Annihilators

- $\emptyset$ is the identity for $+$.
 - $R + \emptyset = R$.
- ε is the identity for concatenation.
 - $\varepsilon R = R\varepsilon = R$.
- $\emptyset$ is the annihilator for concatenation.
 - $\emptyset R = R\emptyset = \emptyset$.

- Examples of Language of Regular Expressions:

- $L(\mathbf{01}) = \{01\}$.
- $L(\mathbf{0^*}) = \{\varepsilon, 0, 00, 000, \dots\}$.
- $L((\mathbf{0+10})^*(\varepsilon+\mathbf{1}))$ = all strings of 0's and 1's without two consecutive 1's.

Regular expressions

1. $\emptyset + R = R$

2. $\emptyset R = R\emptyset = \emptyset$

3. $\varepsilon R = R \varepsilon = R$

4. $\varepsilon^* = \varepsilon$

5. $R + R = R$

6. $R^*R^* = R^*$

7. $RR^* = R^*R$

8. $(R^*)^* = R^*$

9. $\varepsilon + RR^* = R^*$

10. $(PQ)^*P = P(QP)^*$

11. $(P+Q)^* = (P^*Q^*)^* = (P^*+Q^*)^*$

12. $(P+Q)R = PR + QR$

13. $R(P+Q) = RP + RQ$

Examples

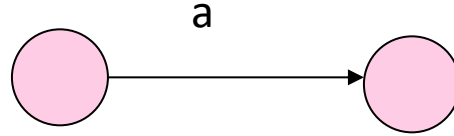
- **$L = \{w \mid \text{length of } w = 2\}$ over $\Sigma = \{a, b\}$**
 - $(a + b).(a + b)$
- **$L = \{w \mid \text{length of } w \geq 2\}$ over $\Sigma = \{a, b\}$**
 - $(a + b).(a + b).(a + b)^*$
- **$L = \{w \mid \text{length of } w \text{ is even}\}$ over $\Sigma = \{a, b\}$**
 - $((a + b).(a + b))^*$
- **$L = \{w \mid \text{no. of } a\text{'s in } w = 2\}$ over $\Sigma = \{a, b\}$**
 - $b^*ab^*ab^*$
- **$L = \{w \mid \text{no. of } a\text{'s in } w \geq 2\}$ over $\Sigma = \{a, b\}$**
 - $b^*ab^*a(a + b)^*$
- **$L = \{w \mid \text{no. of } a\text{'s in } w \text{ is even}\}$ over $\Sigma = \{a, b\}$**
 - $(b^*ab^*ab^*)^* + b^*$

Examples

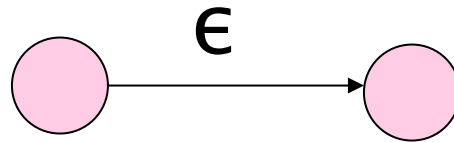
- **Language = $\{w \mid w \text{ contains a single '1' over } \{0,1\}\}$**
 - RE: 0^*10^* (any number of 0s, followed by one 1, then any number of 0s)
- **Language = $\{w \mid w \text{ contains '1' at every odd position and total length is odd}\}$**
 - RE: $1((0+1)1)^*$
 - Logic: Starts with 1 (first odd position), then repeats pairs where the second symbol in the pair (the next odd position) is always 1

Regular Expression to ϵ -NFA

- Symbol **a**:



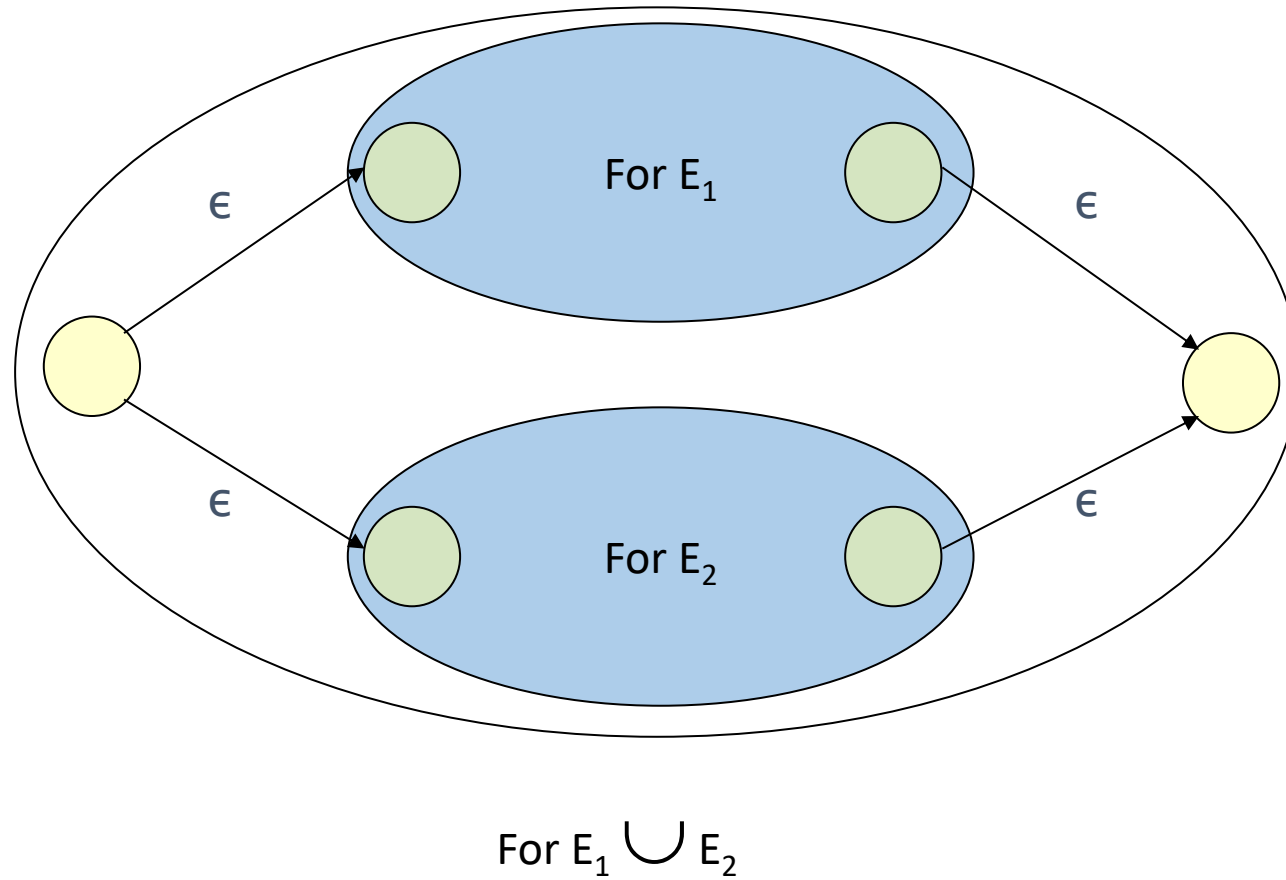
- ϵ :



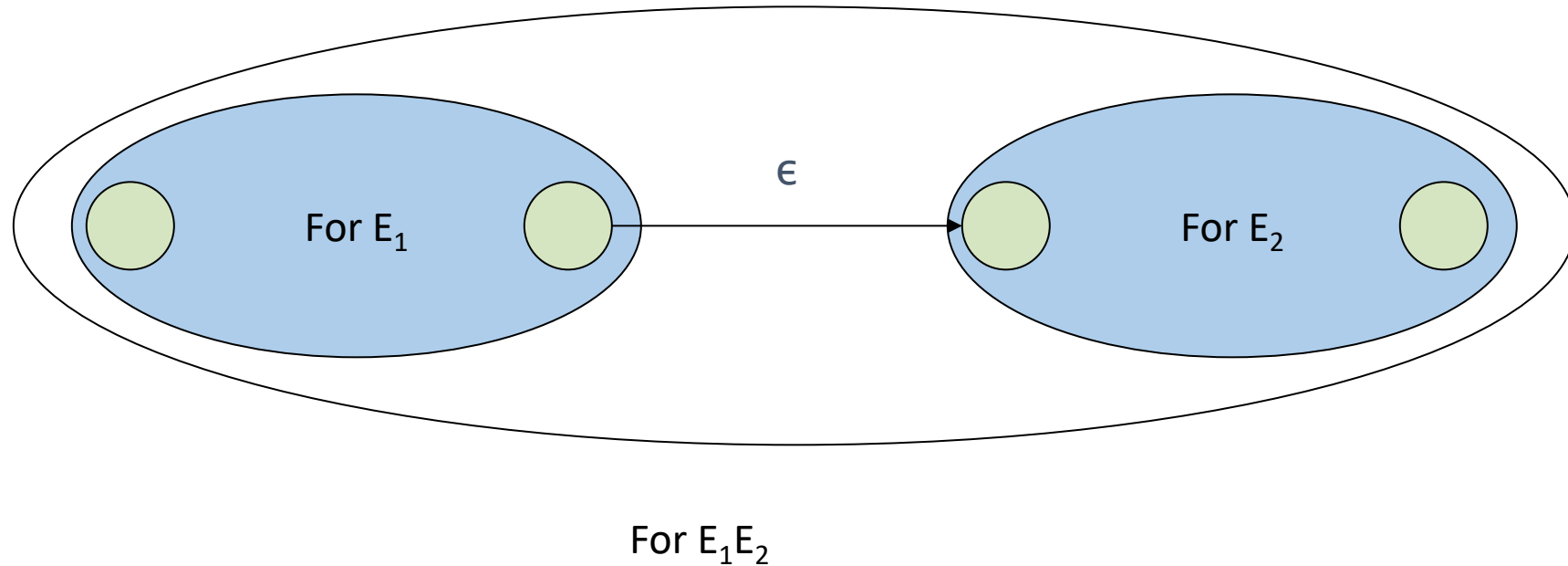
- $\emptyset$:



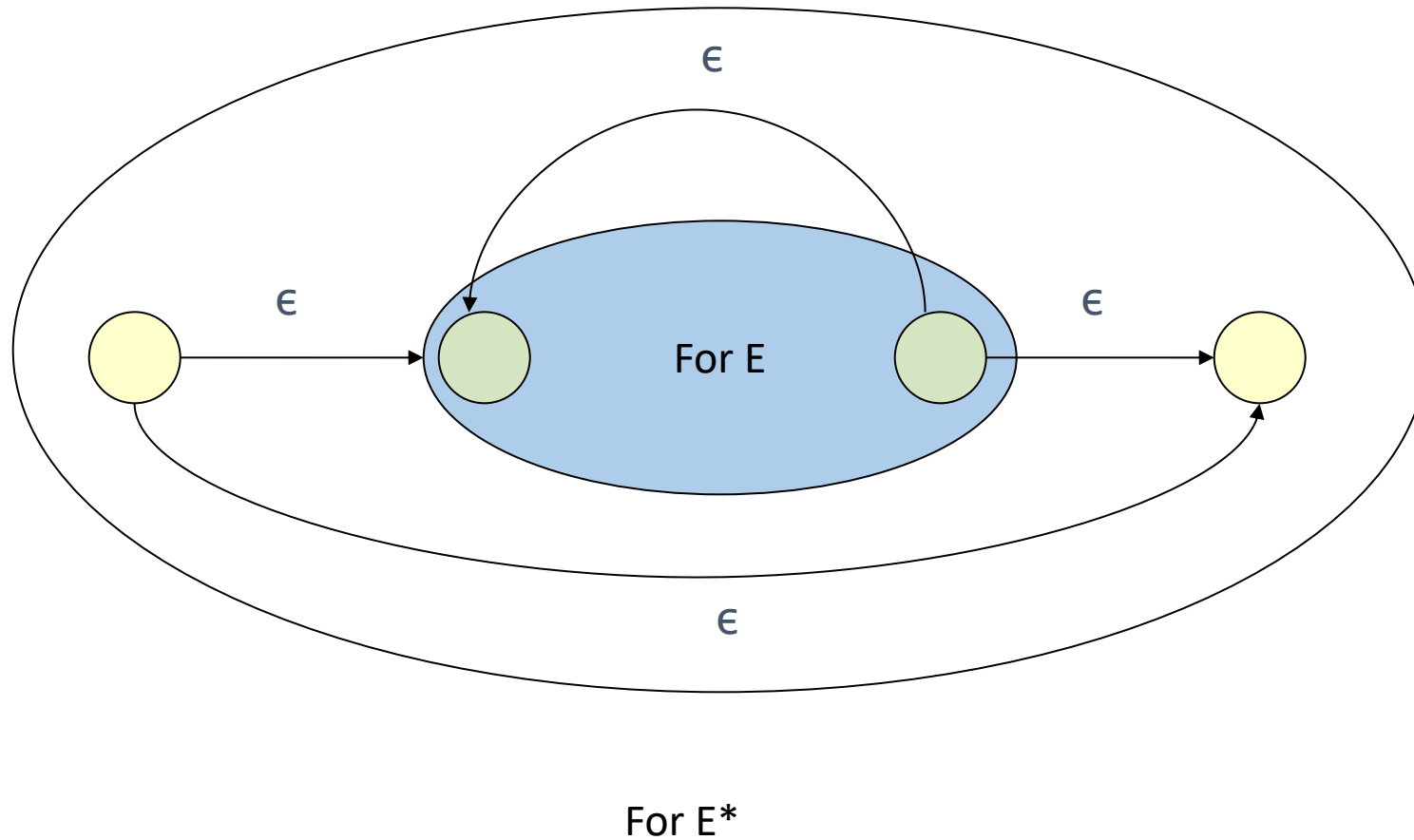
RE to ϵ -NFA: Induction 1 – Union



RE to ϵ -NFA: Induction 2 – Concatenation

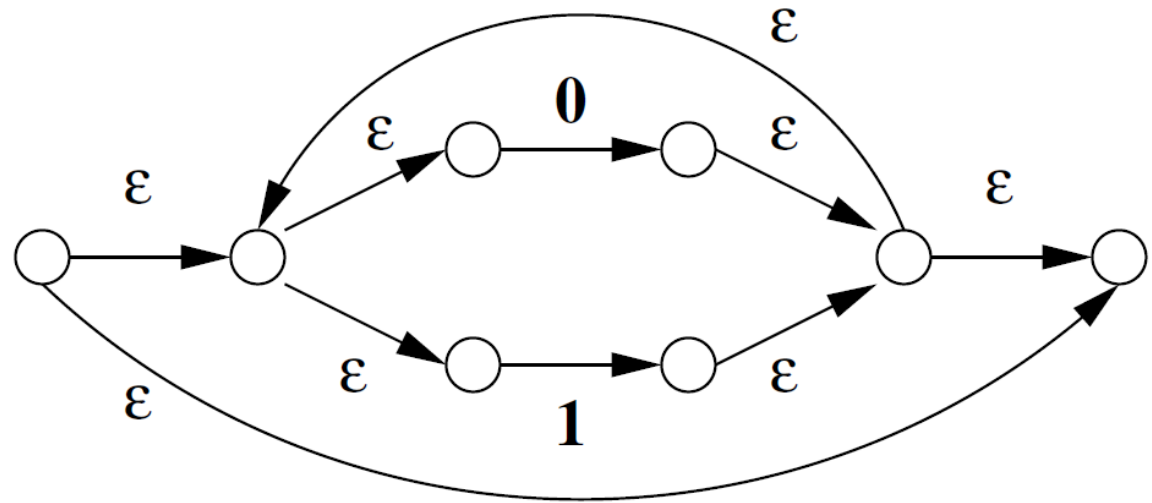
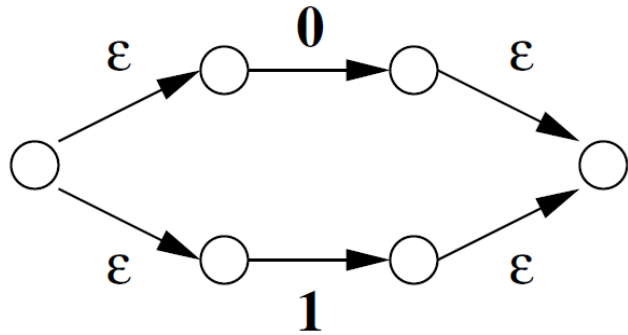


RE to ϵ -NFA: Induction 3 – Closure



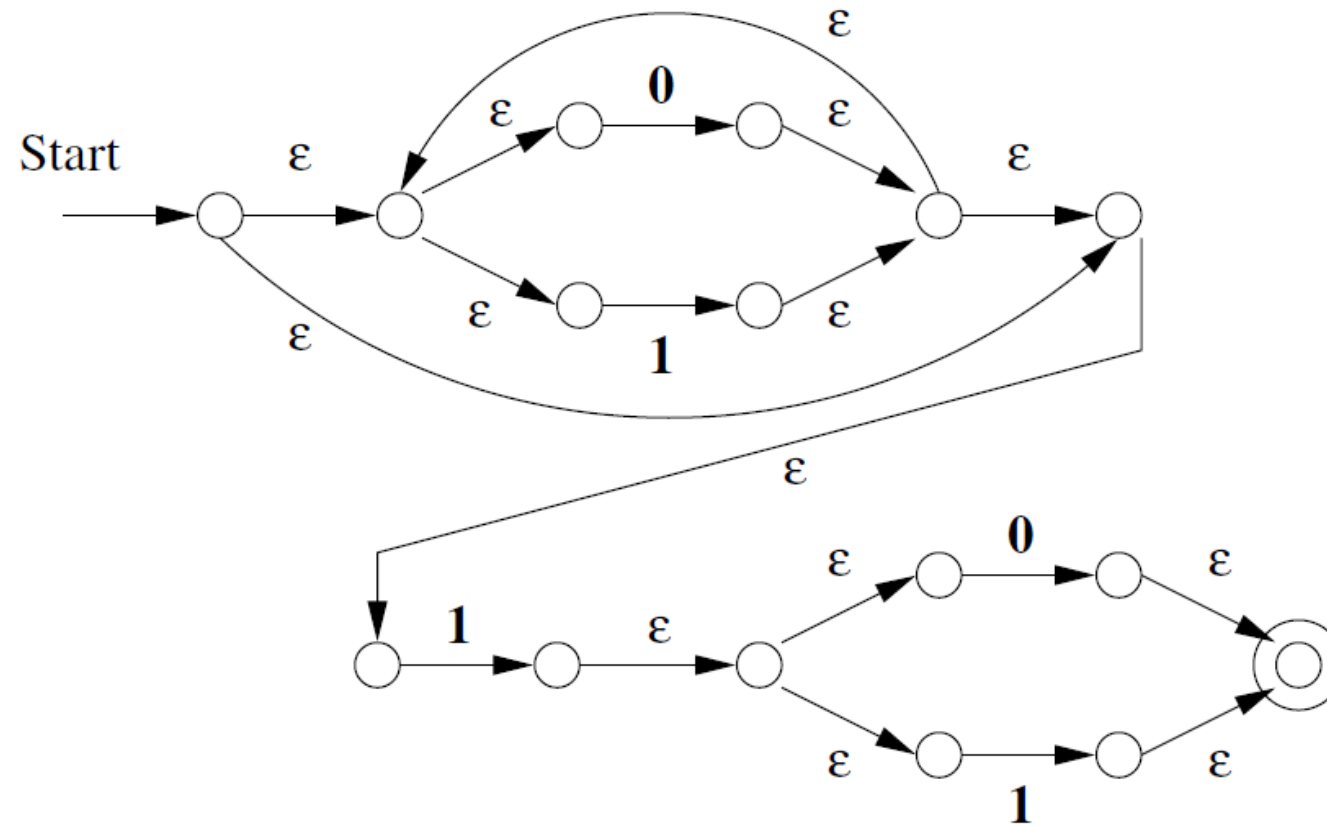
RE to ϵ -NFA: Example

- Convert the regular expression $(0+1)^*1(0+1)$ to an ϵ -NFA



RE to ϵ -NFA: Example

- Convert the regular expression $(0+1)^*1(0+1)$ to an ϵ -NFA



Arden's Theorem

- *Let P and Q be two regular expressions. If P does not contain null string then $R = Q + RP$ has a unique solution $R = QP^*$*

- Proof:

$$\begin{aligned} R &= Q + (Q + RP)P \text{ [After putting the value } R = Q + RP\text{]} \\ &= Q + QP + RPP \end{aligned}$$

Putting the value of R recursively again and again–

$$R = Q + QP + QP^2 + QP^3 \dots$$

$$R = Q (\epsilon + P + P^2 + P^3 + \dots)$$

$$R = QP^* \text{ [As } P^* \text{ represents } (\epsilon + P + P^2 + P^3 + \dots) \text{]}$$

Hence, proved.

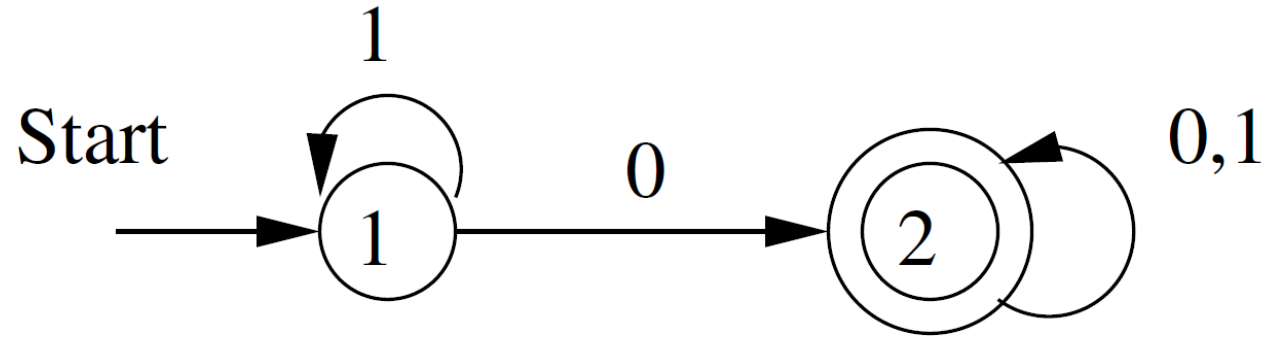
Converting DFA to Regular Expression

Step 1: Construct algebraic equations for each state q_i as follows:

$$q_i = q_j \cdot a \text{ such that } \delta(q_j, a) = q_i$$

Step 2: Add ϵ if q_i is start state

Step 3: Solve the equations using Arden's Theorem to get the solution for final state



$$q_1 = q_1 1 + \epsilon$$

$$q_2 = q_2 1 + q_2 0 + q_1 0$$

$$q_1 = q_1 1 + \epsilon$$

$$q_1 = \epsilon + q_1 1$$

By Arden's Theorem:

$$R = q_1$$

$$Q = \epsilon$$

$$P = 1$$

$$q_1 = R = \epsilon.1^* = 1^*$$

$$q_2 = q_2 1 + q_2 0 + q_1 0$$

$$q_2 = q_2(1 + 0) + 1^*0$$

$$q_2 = 1^*0 + q_2(1 + 0)$$

By Arden's Theorem:

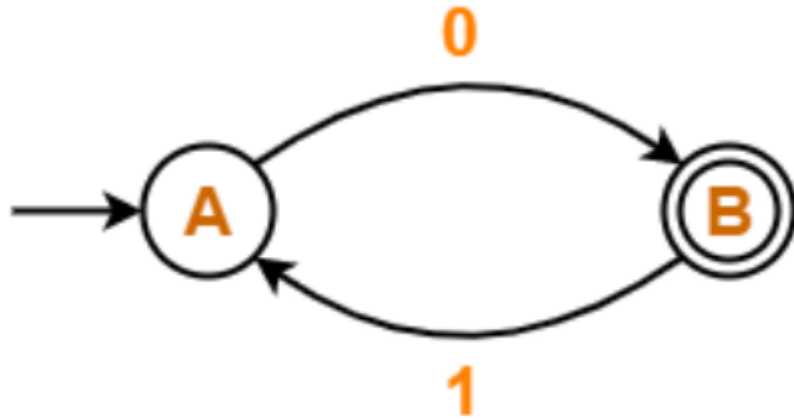
$$R = q_2$$

$$Q = 1^*0$$

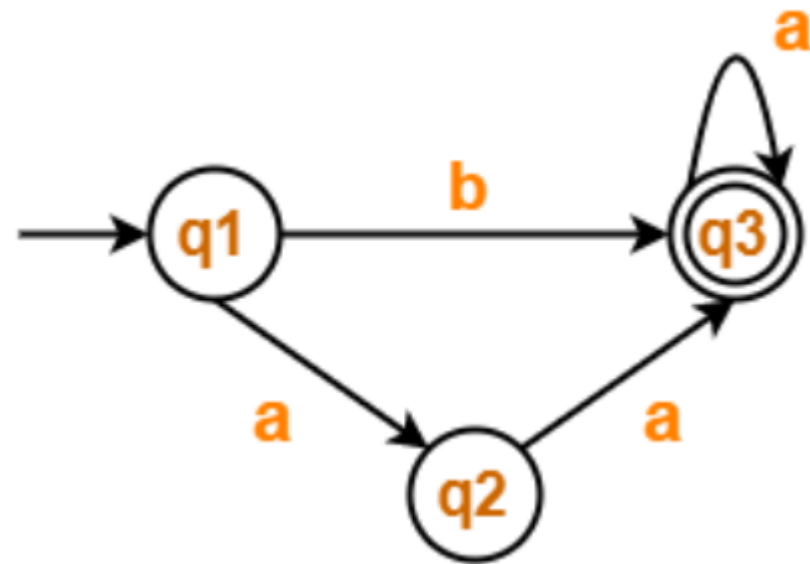
$$P = (1 + 0)$$

$$q_2 = R = 1^*0(1 + 0)^*$$

Examples



$(01)^*0$



$(b+aa)a^*$